Intersection counts and orthogonal graph codes

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Attribution and scope. Counting common neighbours in an association scheme is classical. For uniform subset graphs the count below is Proposition 3 of Key and Rodrigues, *LCD codes from adjacency matrices of graphs*, manuscript dated 4 August 2017 (F017). Their examples and classifications are recovered here from one scalar test and the common orthogonal projection theorem. We also give uniform proofs for the Hamming examples in their conclusion, correcting the stated complement-dual identity for even binary cubes. No historical novelty or resolution of an open conjecture is claimed. Dimensions and distances quoted from other papers in F017 require their own proofs or explicitly retained inputs; the scalar test alone does not prove them.

Write $C_M = \operatorname{im} M$ for a symmetric matrix M over $\mathbb{F}_p$. Let A be the adjacency matrix of a regular simple graph on N vertices, of degree ν, and put $B = J - I - A$. Following F017, call C_A RLCD when $C_A = C_{A+I}^{\perp}$, and CLCD when $C_A = C_B^{\perp}$. Suppose ordered pairs of vertices are partitioned into classes, indexed by j, on which both the adjacency indicator a_j and the number i_j of common neighbours are constant. Include the diagonal class among them.

Theorem 1. *Put $P_j = i_j + a_j$ and $\bar{\nu} = N - 1 - \nu$. Then:*

1. *C_A is RLCD if and only if $P_j = 0$ in $\mathbb{F}_p$ for every j.*
2. *$C_A \perp C_B$ if and only if $\nu - P_j = 0$ for every j. In this event both codes are LCD. They are complementary if and only if $(\nu, \bar{\nu}) \neq (0, 0)$ in the field; otherwise $\mathbb{F}_p^N = C_A \perp\!\!\oplus C_B \perp\!\!\oplus \langle \mathbf{1} \rangle$.*
3. *C_B is RLCD if and only if $\bar{\nu} - \nu + P_j = 0$ for every j.*

Proof. The entries of $A^2 + A$ are P_j. Thus their vanishing is exactly $A(A + I) = 0$. The matrices $-A$ and $I + A$ are then complementary symmetric idempotents. Conversely the asserted duality implies that all their row inner products vanish. Next $AB = AJ - A - A^2 = \nu J - A - A^2$, proving the orthogonality test. The remaining claims in part 2 are the common projection theorem, applied to this actual pair of matrices; no condition on their complex spectra is needed. Finally $B(B + I) = B(J - A) = \bar{\nu} J - AB$, which gives part 3 by the same reflexive argument. □

Uniform subset graphs. Let $n \geq 2k + 1$, $k \geq 2$ and $0 \leq r < k$. The vertices of $\Gamma(n, k, r)$ are the k-subsets of an n-element set; adjacency means intersection of size r. For two vertices meeting in j elements, the common-neighbour count is

$$i_j = \sum_a \binom{j}{a} \binom{k-j}{r-a}^2 \binom{n-2k+j}{k-2r+a}, \qquad 0 \leq j \leq k. \tag{1}$$

Here binomial coefficients with invalid lower argument are zero. To prove the formula, partition the ground set into the intersection, the two differences and the outside. A common neighbour chooses a elements from the intersection, $r - a$ from each difference and the remaining $k - 2r + a$ outside. These choices are disjoint and exhaustive. In particular $i_k = \nu = \binom{k}{r}\binom{n-k}{k-r}$ and $P_j = i_j + \mathbf{1}_{j=r}$, so Theorem 1 applies directly.

For the Johnson graph $r = k - 1$, the only nonzero counts off the diagonal are $i_{k-2} = 4$ and $i_{k-1} = n - 2$. Thus RLCD holds exactly for $p = 2$ and odd n. Under those conditions $\nu = 0$, so CLCD holds exactly when $\binom{n}{k}$ is even. For $k \geq 3$, complement orthogonality itself forces $\nu = 0$ from $P_0 = 0$; then $P_{k-2} = 4$ and $P_{k-1} = n - 1$ force the same characteristic and parity. For $k = 2$, subtraction of the two off-diagonal tests yields $n = 5$ in the field, while $\nu = 4$ gives $2(n - 2) = 4$, hence characteristic two and odd n. The binary dimensions and minimum words follow from the separate simplex boundary theorem, not from this count alone.

The domain qualification matters. F017 Corollary 1(1) does not state it explicitly. For example $J(4, 3) = K_4$ is RLCD in characteristic three. More generally the adjacency matrix of $J(n, n-1) = K_n$, for $n \geq 3$, is $J - I$, so $(J - I)J = (n - 1)J$: it is RLCD exactly when $p \mid (n - 1)$. Its complement is empty, so it is CLCD exactly when $J - I$ is invertible, equivalently $p \nmid (n - 1)$. Thus the unrestricted source exclusion is false. Formula (1) itself remains valid for $n \geq k$ when only realized intersections $\max(0, 2k - n) \leq j \leq k$ are tested; the table below retains its stated $n \geq 2k + 1$ domain.

For a Kneser graph ($r = 0$), formula (1) reduces to $i_j = \binom{n-2k+j}{k}$. If $n = 2k+1$, then $i_0 = 0$ and $P_0 = 1$, so the code is never RLCD. For $k \geq 3$ it is not CLCD either: complement orthogonality would force both $\nu = P_0 = 1$ and $\nu = i_1 = 0$. For $k = 2$ there is an exception. The Petersen graph has $(N, \nu) = (10, 3)$, $P_0 = 1$, $P_1 = 1$ and $P_2 = 3$. Complement orthogonality holds exactly in characteristic two, where $\nu = 1$ and $\bar{\nu} = 0$; hence this code is CLCD. F017 Corollary 1(2) states the exclusion for every k. Its CLCD assertion requires this endpoint correction. Its RLCD exclusion is valid for every $k \geq 2$.

k	r	RLCD	CLCD
2	0	$p = 2$, $n \equiv 3 \pmod 4$	$p = 2$, $n \equiv 1 \pmod 4$
3	0	$p = 2$, $n \equiv 1 \pmod 4$; $p = 3$, $n \equiv 5 \pmod 9$; $p \geq 5$, $n \equiv 5 \pmod p$	$p = 2$, $n \equiv 1 \pmod 4$; $p = 3$, $n \equiv 2, 8 \pmod 9$; $p \geq 5$, $n \equiv 5 \pmod p$
4	0	$p = 2$, $n \equiv 7 \pmod 8$	$p = 2$, $n \equiv 3 \pmod 8$
3	1	$p = 2$, $n \equiv 0 \pmod 4$	$p = 2$, $n \equiv 0, 1 \pmod 4$
4	1	$p = 2$, $n \equiv 1 \pmod 4$; $p = 3$, $n \equiv 6 \pmod 9$	$p = 2$, $n \equiv 1 \pmod 8$; $p = 3$, $n \equiv 6 \pmod 9$

Table 1: Complete parameter conditions for the five families specified in F017, with $n \geq 2k + 1$. Each entry is an if and only if statement.

Completeness of the arithmetic classification. Here is a finite exact certificate for Table 1, including the exclusion of all other primes. For each fixed (k, r), regard the P_j and $\nu - P_j$ as polynomials in n over $\mathbb{Q}$ using (1). For each list $f_0, \ldots, f_k$, the accompanying program records polynomial coefficients b_j and a monic polynomial g such that

$$\sum_j b_j f_j = g, \qquad g \mid f_j \quad \text{for all } j.$$

Both assertions are verified by exact polynomial arithmetic. Let D be the product of a common denominator of all coefficients of all b_j and a common denominator of all coefficients of all f_j. For $p \nmid D$, the simultaneous congruences are equivalent to $g(n) = 0$. In the two $k = 3, r = 0$ lists, $g = n - 5$; in all other lists, $g = 1$. The resulting exceptional primes, including primes subsequently

excluded by the finite tests, are

(k,r)	P_j	$\nu - P_j$
$(2,0)$	2	2
$(3,0)$	$2,3$	$2,3$
$(4,0)$	$2,3$	$2,3,17$
$(3,1)$	$2,3,5$	$2,3,23$
$(4,1)$	$2,3,5$	$2,3,5.$

For an exceptional prime choose the least power $p^a > k$. For $0 \le b \le k$, the identity $(1+X)^{p^a} = 1 + X^{p^a}$ over $\mathbb{F}_p$ proves $\binom{m+p^a}{b} = \binom{m}{b}$ modulo p for nonnegative m. Consequently all the counts, the degree and $\binom{n}{k}$ have period p^a modulo p in the domain $n \ge 2k+1$. Testing one admissible lift of every residue modulo p^a is therefore exhaustive, not a bounded experiment. The program records these residues and applies the constant-line exclusion in Theorem 1 to obtain CLCD. For $k = 3, r = 0, p \ge 5, n \equiv 5$, the degree is zero but $\binom{n}{3} - 1 \equiv 9 \ne 0$, so no further exception occurs. This proves the table from its exact arithmetic certificate.

Hamming graphs. Let $H_k(d,q)$ have vertex set $\{0,\dots,q-1\}^d$, with adjacency at Hamming distance k, where $1 \le k \le d$ and $q \ge 2$. If two vertices have distance j, their common-neighbour count is

$$i_j = \sum_a \binom{d-j}{a}(q-1)^a \frac{j!}{b!^2 e!}(q-2)^e, \qquad b = j+a-k, \quad e = 2k-j-2a, \tag{2}$$

where terms with negative factorial arguments are omitted. In their $d-j$ equal coordinates, choose the a positions where a common neighbour differs. Among the j different coordinates, let b agree with the first vertex, b with the second, and e with neither. The distance equations give the stated values of b, e, with $2b + e = j$. Multinomial counting proves (2). Theorem 1 now applies with $N = q^d$, $\nu = i_0 = \binom{d}{k}(q-1)^k$ and $P_j = i_j + \mathbf{1}_{j=k}$.

Corollary 2. *For odd q, every binary distance-k Hamming code is RLCD. For the distance-two binary Hamming graph over $\mathbb{F}_3$, RLCD holds whenever $d \equiv 0 \pmod 3$.*

Proof. Let X_i act as the complete-graph adjacency matrix $J_q - I_q$ in coordinate i and as the identity in all others. Over $\mathbb{F}_2$, odd q gives $X_i^2 = X_i$, and these matrices commute. The distance-k adjacency matrix is $A_k = \sum_{|S|=k} \prod_{i \in S} X_i$. Squaring cancels mixed terms in characteristic two, so $A_k^2 = A_k$ and $A_k(A_k + I) = 0$. For $q = 2, k = 2$, the only possible common-neighbour distances are $0, 2, 4$, with counts $\binom{d}{2}$, $2(d-2)$ and 6, respectively. If $3 \mid d$, these are $0, -1, 0$ modulo three. This is precisely the RLCD test. □

For completeness, the Hamming spectrum used to compute the dimensions comes from the same tensor construction. Over $\mathbb{Q}$, $J_q - I_q$ has eigenvalue $q-1$ on the constant vectors and -1 on their complement. Choosing t nonconstant tensor factors therefore gives

$$\lambda_t = \sum_i (-1)^i (q-1)^{k-i} \binom{t}{i}\binom{d-t}{k-i}, \qquad m_t = \binom{d}{t}(q-1)^t.$$

The common LCD rank lemma sums precisely those multiplicities for which $p \nmid \lambda_t$. In particular the ternary $H_2(6,2)$ and $H_2(9,2)$ codes have dimensions 42 and 342. This rank step is justified by the proved LCD property, rather than an assumption of diagonalizability after reduction modulo p.

Theorem 3. *Over $\mathbb{F}_2$, the distance-k binary Hamming adjacency matrix satisfies $A_k^2 = \binom{d}{k} I$. For positive even d, the code of the ordinary cube $Q_d = H_1(d, 2)$ is self-dual of dimension 2^{d-1}. Its complement adjacency matrix is invertible, so the code is not the dual of the complement code.*

Proof. Index coordinates by $\mathbb{F}_2^d$. Write T_s for translation by s. Then $A_k = \sum_{\mathrm{wt}(s)=k} T_s$. The translations commute, each satisfies $T_s^2 = I$, and cross terms cancel in the square, proving the first assertion. Write M_t for the adjacency matrix of the ordinary cube Q_t. Order cube vertices by their last coordinate. If d is even,

$$M_d = \begin{pmatrix} M_{d-1} & I \\ I & M_{d-1} \end{pmatrix}, \qquad M_{d-1}^2 = I.$$

The kernel consists exactly of $(x, M_{d-1}x)$, so both nullity and rank are 2^{d-1}. Since $M_d^2 = 0$ and M_d is symmetric, its image equals its orthogonal complement. On the other hand, with $B = J + I + M_d$ we have $J^2 = 2^d J = 0$ and $JM_d = M_d J = dJ = 0$, so $B^2 = I$. □

The conclusion of F017 displays the complement-dual identity for Q_6, Q_8, Q_{10}. Direct reading of the formula confirms the complement bar; Theorem 3 disproves that identity and supplies the corrected self-duality statement. The two remaining Hamming examples in that conclusion are instances of Corollary 2 and are retained.

Verification and limits. The required scalar certificate is `check_association_counts.py` and its exact polynomial and residue report. Separate explicit finite graphs test the counting formulas on all vertex pairs in those test graphs; these tests support implementation correctness and are not used to infer the infinite classification. No finite minimum-distance assertion is inferred from this manuscript. The characteristic-polynomial rank lemma and the general projection theorem are shared dependencies, each charged in the common theory when this work is measured.